

ROAD GUARDIAN AI - SYSTEM PORTFOLIO

Urban Digital Twin, AI Perception, & Traffic Rerouting Platform

Platform Version: 2.0.0 (Production Core)

Date Compiled: 2026-08-29 | Target: Municipal & Hackathon Review

1. Executive Summary

Road Guardian AI is an end-to-end smart city infrastructure monitoring platform designed to detect road hazards (potholes), evaluate local traffic conditions, analyze image report authenticity, and simulate regional detours using WebGL spatial maps. The platform serves both citizens (via hazard reporting) and municipal PWD/NHAI authorities (via analytics & detour planning tools) to accelerate city repairs and ensure road safety.

2. Core Sub-Systems & Architectural Layers

- AI Perception Scanner:

Powered by a custom-trained PyTorch YOLOv8 model targeting real-time pothole identification with bounding-box annotations.

- MapTiler Digital Twin:

WebGL-powered vector map visualizing dynamic database records, reported pothole coordinate pins, and color-coded road risk segment corridors.

- Authenticity Check Engine:

A 7-stage computer vision verification pipeline (checking EXIF data, GPS geotags, pHash duplicate matches, screen photo Fourier moire patterns, JPEG ELA splicing, and noise spectrum AI details) to reject fake citizen uploads.

- Traffic Simulation Engine:

Allows municipal admins to trigger segment closures and simulate citywide capacity/volume redistribution on alternate routes.

- Risk Calculator Engine:

Calculates a 0-100 multi-factor risk score by checking hazard severity, vehicle speeds, weather condition, traffic, and hospital/school proximity.

3. Technology Stack & Dependencies

Layer / Dependency	Technology Choice	Role in Platform
Backend Framework	FastAPI (Python)	High-performance async APIs & static routers
Computer Vision Models	YOLOv8 & OpenCV	Pothole scanning & image manipulation checks
Database Engine	SQLite / MySQL	Unified storage for report logs and telemetry
Mapping SDK	MapTiler GL JS	Dynamic WebGL maps with vector styling layers
Frontend UI Framework	React 18 & Tailwind	Dashboard panels & admin control portal
Report Generation	FPDF (Python)	Executive municipal PDF audits compilation

4. Security, Deduplication & Maintenance

- Spatial-Temporal Deduplication: Rejects identical hazard logs submitted within 10 meters and 2 hours of an active report to prevent spam.
- Admin Purge Controller: A passcode-protected municipal maintenance panel to securely wipe all database records and clean stored files.

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3. Live Sensor Geolocation: Browser checks query high-precision GPS sensors directly and output Degrees, Minutes, and Seconds (DMS) coordinates down to the decimal second.